

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I M. Vishnu Vardhan Reddy (192572062) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Application Based Questions

- 1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.**

Cloud **auto-scaling** automatically increases or decreases computing resources based on application demand. During seasonal sales or promotional events, user traffic can increase suddenly. Auto-scaling launches additional virtual machines or containers to handle the increased workload and removes them when traffic decreases, reducing unnecessary costs. **Load balancing** distributes incoming user requests evenly across multiple servers. This prevents any single server from becoming overloaded and ensures that all servers are used efficiently. If one server fails, the load balancer automatically redirects traffic to healthy servers, improving system availability.

This combination significantly reduces downtime, improves response time, and provides a smooth user experience even during heavy traffic. **Monitoring tools** such as CloudWatch, Azure Monitor, or Google Cloud Monitoring continuously monitor CPU usage, memory utilization, network traffic, and response time. When resource utilization reaches a predefined threshold, monitoring tools automatically trigger the auto-scaling process.

Example: During Amazon's Great Indian Festival or Flipkart's Big Billion Days sale, millions of users visit the website simultaneously. Auto-scaling automatically adds new servers to manage the increased demand, while load balancers distribute customer requests evenly, ensuring fast page loading and uninterrupted shopping.

2. **An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.**

Cloud Storage for Backup Data

Cloud storage is widely used to store large amounts of backup data because it provides high durability, availability, and scalability. Cloud providers maintain multiple copies of data across different data centers and geographic regions. Even if one storage device or data center fails, the backup data remains available from another location.

Encryption protects sensitive backup data both during transmission and while stored in the cloud. Only authorized users with proper authentication can access the encrypted data. **Access control** mechanisms such as Role-Based Access Control (RBAC) and Multi-Factor Authentication (MFA) prevent unauthorized access and improve security.

Compared to traditional storage systems, cloud storage eliminates the need to purchase expensive hardware, maintain storage devices, or perform regular upgrades. Organizations pay only for the storage capacity they actually use, making cloud storage a cost-effective solution.

Example: If a company's primary data center is damaged due to a fire or flood, cloud backups can be restored quickly to a new server, allowing business operations to continue with minimal downtime. This is an example of **Disaster Recovery (DR)**.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

A **Content Delivery Network (CDN)** is a network of distributed servers located in different geographic regions. Instead of serving content from a single central server, CDNs store copies of web pages, images, videos, and application files on **edge servers** that are located closer to users.

When a user requests content, the CDN delivers it from the nearest edge server, reducing network latency and improving loading speed. This decreases the distance data must travel and reduces congestion on the main server. Cloud providers maintain data centers worldwide, ensuring high availability and reliable performance even during peak traffic.

CDNs also reduce bandwidth usage on the origin server, improve scalability, and protect applications from traffic spikes and certain cyberattacks.

Example: Netflix stores video content on CDN edge servers across multiple countries. When users watch movies, the video is streamed from the nearest server, reducing buffering and improving video quality. Similarly, online multiplayer games use CDNs to reduce latency and provide a smoother gaming experience.

- 4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.**

Organizations such as banks and financial institutions must comply with strict government and industry regulations when storing customer data in the cloud. Cloud providers support compliance by offering internationally recognized security certifications, regular security audits, compliance reports, and built-in security controls.

Logging records every user activity, login attempt, and system event. **Monitoring** continuously analyzes system behavior to detect suspicious activities and security threats. **Encryption** protects sensitive information while it is stored and during transmission across networks, ensuring that unauthorized users cannot read the data.

To meet **data sovereignty** requirements, cloud providers allow organizations to store and process data only within approved geographic regions or countries, ensuring compliance with local privacy laws.

Example: A bank encrypts customer account information and transaction records before storing them in a cloud data center located within the country. Logging and monitoring continuously track all access to financial records, helping the organization satisfy regulatory compliance and detect fraudulent activities.

- 5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.**

Provide an example of a microservices-based application.

Containers are lightweight virtualization technologies that package an application together with all its libraries, dependencies, and configuration files. Unlike virtual machines, containers share the host operating system, making them consume fewer resources, start much faster, and support higher application density.

Kubernetes is a container orchestration platform that automates deployment, scaling, load balancing, service discovery, rolling updates, and automatic recovery of containerized applications. It continuously monitors containers and restarts failed containers automatically, improving application reliability.

Containers provide excellent portability, allowing applications to run consistently across private clouds, public clouds, and on-premises infrastructure without requiring significant modifications. This simplifies application deployment across different cloud providers and

avoids vendor lock-in.

Example: An e-commerce application built using a microservices architecture may have separate containers for user authentication, product catalog, payment processing, inventory management, order processing, and notification services. Kubernetes automatically manages these containers, scales individual services based on demand, and ensures uninterrupted application availability during traffic spikes.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity